

Total No. of Questions : 8]

SEAT No. :

PE-4259

[Total No. of Pages : 3

[6582]-31

S.E. (Electronics and Computer Engineering)
SYSTEM PROGRAMMING AND OPERATING SYSTEMS
(2019 Pattern) (Semester - IV) (204207)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Assume suitable data, if necessary.

- Q1)** a) Draw and explain architecture of Multitasking Operating Systems. [6]
b) Consider the following processes arrival time and burst time are as shown. Calculate average waiting time and average turnaround time using FCFS scheduling algorithm. [6]

Process	Arrival time	Burst time
P1	02	02
P2	00	01
P3	02	03
P4	03	05
P5	04	05

- c) Define Process. Draw process state transition diagram in operating system and explain different process states in detail. [8]

OR

- Q2)** a) In the following example, there are five processes named as P1, P2, P3, P4 and P5. Their arrival time and burst time are given in the table below. Calculate average waiting time and average turnaround time for SJF (Non-preemptive) scheduling algorithm. [6]

Process	Arrival time	Burst time
P1	02	06
P2	05	02
P3	01	08
P4	00	03
P5	04	04

- b) Draw and explain architecture of Batch Operating Systems. [6]
c) Draw and explain Microkernel Operating System. Differentiate between Microkernel and Monolithic Kernel. [8]

P.T.O.

- Q3)** a) Explain any two Classical Problems of Synchronization in detail. [8]
b) What is semaphore? Explain different types of Semaphore. State the advantages of Semaphore. [8]

OR

- Q4)** a) What is dead lock in operating system? What are the three ways to handle deadlock explain it with example. [8]
b) What is Mutual Exclusion? Explain any three mutual exclusion techniques in detail. [8]

- Q5)** a) Explain the concept of Paging and Segmentation. Differentiate between Paging and Segmentation. [8]
b) Explain the terms Shared Memory and Message passing for IPC. Give the difference between shared memory and message passing. [8]

OR

- Q6)** a) i) Explain concept of paging with suitable diagram.
ii) Calculate the number of page faults and page hits for the page replacement policies FIFO for the given reference string 1, 3, 0, 3, 5, 6, 3 with 3 page frames. [8]
b) What is Fragmentation? Explain Internal fragmentation and External fragmentation and differentiate between them. [8]

- Q7)** a) Differentiate between Programmed I/O and Interrupt Driven I/O. [6]
b) What is I/O buffering? Explain different types of I/O buffering. [6]
c) Explain the following File allocation methods [6]
i) Contiguous File allocation
ii) Linked File Allocation

OR

- Q8) a)** Explain the following Directory structure in OS [6]
- i) Single-level directory
 - ii) Two-level directory
- b) Consider the following disk request sequence for a disk with 200 tracks
Request
sequence = {176, 79, 34, 60, 92, 11, 41, 114}
Head pointer is starting at 50 and moving in left direction. Find the number of head movements (total seek length) in cylinders using FCFS scheduling. [6]
- c) Explain file management concept. What are the functions of file management in operating system? [6]

